Session 5: Object-Oriented Programming

Trainer Reference Guide

Purpose and outcomes

Model cohesive data and behavior with classes/objects, __init__, methods, class attributes, __str__, encapsulation, inheritance, composition, and simple polymorphism.

Examples and exercises

- 01_classes_objects.ipynb creates User instances and guides active-state behavior.
- models/user.py encapsulates normalized user state; Employee(User) adds employee ID/department; Notification and EmailNotification share a send interface.
- 02_encapsulation.py changes ticket status only through a validated method.
- 03_inheritance_composition.ipynb composes TicketService with a replaceable notifier.
- exercise_01_user_class.py is guided. exercise_02_employee.py independently applies inheritance/super. challenge_notifications_tickets.py combines required classes.
- ticket_system.py is Project 5: Ticket, Agent, polymorphic Notification, and a composed TicketManager.

Trainer key points specific to this module

Use ?is-a? for inheritance and ?uses-a/has-a? for composition. Show why ‘_status‘ signals controlled access. Watch forgotten ‘self‘, mutable class state, missing ‘super()‘, and inheritance used only to reuse code.

Portal connection and evidence

UserService composes repository and email service. Learners can swap a notifier without editing ticket logic.

Delivery pattern

1. Explain the idea briefly and ask learners to predict a result.
2. Run the simple example, then connect it to an employee/user workflow.
3. Pair learners for the guided task; require an individual attempt next.
4. Use the challenge as the cumulative project increment.
5. Review the quiz and ask learners to explain, not only show, their code.

Trainer key points

- Keep live coding visible and deliberate. Normalize reading tracebacks.
- Ask what type and value each expression produces before running it.
- Let TODO files remain student-owned; open solutions only during review.
- Check boundary and failure cases, not only the happy path.
- End by tracing how the topic appears in the final portal.